

Abrisam Hafiz Azhar

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PERSONAL STATEMENT

Third-year MEng Mechanical Engineering student at the University of Sheffield (First Class Honours, expected 2027) with research experience in railway engineering and hands-on design leadership across multidisciplinary projects. Completed FTIR spectroscopy and tribometry research to detect friction modifiers on railway tracks and delivered MATLAB-based analysis outputs for industry-relevant insights. Led a sub team for a wind turbine build (750 W) and won the Aegis Engineering Design Competition. Seeking a Summer Internship Role, applying strengths in thermodynamics, fluid dynamics, engineering design, and test-informed problem solving.

EDUCATION

University of Sheffield, UK | MEng Mechanical Engineering | Sep 2023 – July 2027 (Expected)

Current Standing: First Class Honours (1st)

Awards: Sir Harold West Award Winner*, Engineering Excellence Scholarship, SURE Research Scholarship

Relevant Modules: Manufacturing, Structural Dynamics, Fluid Dynamics, Thermodynamics

*Awarded annually to eight first-year undergraduates in the Faculty of Engineering, in recognition of academic and personal promise.

TECHNICAL HIGHLIGHTS

Engineering Design and Build: Fusion 360, prototyping, 3D printing, laser cutting, composite fabrication

Simulation and Analysis: FEA, CFD, MATLAB (data processing and scripting)

Testing and Instrumentation: FTIR spectroscopy, tribometry, 3D scanning, quality control

Automation and Control: Arduino, Universal Robots (6-axis) programming

PROJECTS

FTIR Analysis of Friction Modifiers on Railway Tracks | Summer Research (6 weeks) | Jun – Aug 2025

- Planned and executed 5 experiments to detect friction modifiers (FM) on railway tracks using a handheld FTIR (Fourier Transform Infrared) spectroscopy device and a tribometer.
- Processed and analysed 20 datasets in MATLAB, producing 6 final infrared absorption spectra and interpreting correlations between surface chemistry and friction behaviour.
- Documented methodology, results, and limitations clearly to support ongoing industry-facing research workstreams.

EAPC Gearbox (Second Year Design Project) | Sep 2024 - May 2025

- Designed and built a 250 W gearbox to convert a standard bicycle into an electric-assisted pedal cycle (15.5 mph target).
- Designed prototype components in Fusion 360 and built control integration using Arduino, 3D printing, and laser cutting.
- Conducted FEA on gearbox casing and implemented design changes that reduced material use by 30% while maintaining structural integrity, supported by simulation review and prototype testing.

Wind Turbine Project (Blades Sub Team Lead) | Oct 2024 – Present

- Led a team of five (three sub-teams) to design and manufacture a 750 W wind turbine for Whirlow Hall Farm, coordinating build plan, deadlines, and stakeholder updates.
- Modelled turbine blades in Fusion 360 and led fabrication of three 1.2 m wooden blades, including balance test planning and surface finishing workflow.
- Implemented 3D scanning to compare manufactured blades against CAD geometry, enabling early defect detection and reducing rework during balancing and integration.

Mind The Gap, Aegis Engineering Design Competition (Winner) | Feb 2024

- Led a multidisciplinary team to develop a passively actuated ramp supporting up to 200 kg, within one week.
- Designed and fabricated a functional prototype using Fusion360, laser cutting, and 3D printing in 2 days.
- Secured 1st place out of 15 teams for creativity, feasibility, and execution.

University of Sheffield Robotic Arm Competition | Jan – Mar 2025

- Trained to program and operate a 6-axis Universal Robots robotic arm, gaining practical experience in industrial automation and robotic control systems.

WORK EXPERIENCE

Kitlocker.com – Production Staff | Jul - Sep 2024, 2025

- Operated heat press machines for printing 100+ custom apparel daily in high-volume production environment across two summer terms.
- Maintained quality control standards across production runs, ensuring customer specifications were met.
- Supported workflow improvements through coordination with team members to reduce errors and rework.